

CO3-AT1-Case Study

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Course: Fundamentals of Data Science

Case Study 1-Detecting an Unusual Customer

A shopping website finds that the average amount spent by customers is ₹2,000, with a standard deviation of ₹400.

A particular customer spends ₹3,200.

1. Calculate the customer's z-score.
2. Is the customer's spending unusually high compared with the average?
3. What would a negative z-score mean in this context?
4. If another customer has a z-score of -1.5 , what does it tell you?

Solution:

To solve this case study, use the z-score formula:

$$z = \frac{x - \mu}{\sigma}$$

Where:

- x = Customer's spending = ₹3200
- μ = Mean spending = ₹2000
- σ = Standard deviation = ₹400

1. Calculate the customer's z-score

$$\begin{aligned} z &= \frac{3200 - 2000}{400} \\ &= \frac{1200}{400} \\ &= 3 \end{aligned}$$

Answer: $z = 3$

2. Is the customer's spending unusually high?

Yes. A z-score of 3 means the customer's spending is 3 standard deviations above the average. Generally, any z-score greater than +2 or less than -2 is considered unusual.

3. What would a negative z-score mean in this context?

A negative z-score means the customer spends less than the average amount.

For example:

- $z = -1$ means the customer spends 1 standard deviation below the mean.

4. If another customer has a z-score of -1.5 , what does it tell you?

A z-score of -1.5 means the customer's spending is 1.5 standard deviations below the average.

Calculate the spending amount:

$$\begin{aligned}x &= \mu + z\sigma \\x &= 2000 + (-1.5 \times 400) = 2000 - 600 = 1400\end{aligned}$$

Answer:

- The customer spends ₹1,400.
- Their spending is below the average, but it is not unusually low because the z-score is greater than -2 .

Case Study 2-Delivery Time Analysis

A delivery company records delivery time (in minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

- 1. Calculate the mean and median.**
- 2. Find Q1 and Q3.**
- 3. Calculate the IQR.**
- 4. Check for outliers using the $1.5 \times \text{IQR}$ rule.**
- 5. Explain whether the mean or median gives a better description of the typical delivery time.**

Solution:

Given delivery times (in minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

1. Calculate the Mean and Median

Formula for Mean:

$$\text{Mean} = \frac{\sum X}{N}$$

Where:

- $\sum X$ = Sum of all observations
- N = Number of observations

Calculation:

$$\begin{aligned}\sum X &= 25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60 = 326 \\ N &= 10 \\ \text{Mean} &= \frac{326}{10} = 32.6\end{aligned}$$

Mean = 32.6 minutes

Formula for Median:

For an even number of observations,

$$\text{Median} = \frac{\left(\frac{N}{2}\right)^{th} \text{ value} + \left(\frac{N}{2} + 1\right)^{th} \text{ value}}{2}$$

Calculation:

Ordered data: 25, 27, 28, 29, 30, 30, 31, 32, 34, 60

$$\text{Median} = \frac{30 + 30}{2} = 30$$

Median = 30 minutes

2. Find Q1 and Q3

Formula:

- Q_1 = Median of the lower half
- Q_3 = Median of the upper half

Calculation:

Lower half: 25, 27, 28, 29, 30

Median of lower half = 28

$$Q_1 = 28$$

Upper half: 30, 31, 32, 34, 60

Median of upper half = 32

$$Q_3 = 32$$

$$Q_1 = 28$$

$$Q_3 = 32$$

3. Calculate the Interquartile Range (IQR)

Formula:

$$IQR = Q_3 - Q_1$$

Calculation:

$$IQR = 32 - 28 = 4$$

$$IQR = 4$$

4. Check for Outliers using the $1.5 \times IQR$ Rule

Formula:

$$\text{Lower Fence: } Q_1 - 1.5(IQR)$$

$$\text{Upper Fence: } Q_3 + 1.5(IQR)$$

Calculation:

$$1.5 \times 4 = 6$$

$$\text{Lower Fence: } 28 - 6 = 22$$

$$\text{Upper Fence: } 32 + 6 = 38$$

Any value:

- Less than 22
- Greater than 38

is an outlier.

Since

$$60 > 38$$

Outlier = 60 minutes

5. Which is a Better Measure: Mean or Median?

The value 60 is an outlier, which increases the mean from the typical delivery times.

- Mean = 32.6 minutes
- Median = 30 minutes

The median gives a better description of the typical delivery time because it is not influenced by the outlier.

Case Study 3-Two Cities Temperatures

The average daily temperatures (°C) for 7 days are:

City A: 28, 29, 30, 30, 31, 30, 32

City B: 20, 25, 30, 35, 30, 25, 45

1. Calculate the mean temperature for each city.
2. Calculate the variance for each city.
3. Which city has greater temperature variability?
4. Why can two datasets have the same mean but different standard deviations?
5. Which city has more predictable temperatures?

Solution:

Given Data

City A: 28, 29, 30, 30, 31, 30, 32

City B: 20, 25, 30, 35, 30, 25, 45

1. Calculate the Mean Temperature

Formula:

$$\bar{x} = \frac{\sum X}{N}$$

where:

- $\sum X$ = Sum of observations
- N = Number of observations

City A:

$$\sum X = 28 + 29 + 30 + 30 + 31 + 30 + 32 = 210$$

$$\bar{x} = \frac{210}{7} = 30^{\circ}\text{C}$$

Mean of City A = 30°C

City B:

$$\sum X = 20 + 25 + 30 + 35 + 30 + 25 + 45 = 210$$

$$\bar{x} = \frac{210}{7} = 30^{\circ}\text{C}$$

Mean of City B = 30°C

2. Calculate the Variance

Formula (Population Variance):

$$\sigma^2 = \frac{\sum (X - \bar{x})^2}{N}$$

City A:

X	$X - \bar{x}$	$(X - \bar{x})^2$
28	-2	4
29	-1	1
30	0	0
30	0	0
31	1	1
30	0	0
32	2	4

$$\sum (X - \bar{x})^2 = 10$$

$$\sigma^2 = \frac{10}{7} = 1.43$$

Variance of City A = 1.43

City B:

X	$X - \bar{x}$	$(X - \bar{x})^2$
20	-10	100
25	-5	25
30	0	0
35	5	25
30	0	0
25	-5	25
45	15	225

$$\sum(X - \bar{x})^2 = 400$$

$$\sigma^2 = \frac{400}{7} = 57.14$$

Variance of City B = 57.14

3. Which City Has Greater Temperature Variability?

Since

- Variance of City A = 1.43
- Variance of City B = 57.14

Answer: City B has greater temperature variability because its temperatures are spread much farther from the mean.

4. Why Can Two Datasets Have the Same Mean but Different Standard Deviations?

The mean measures the average value, while the standard deviation measures how spread out the data is around the mean.

Although both cities have the same average temperature (30°C), City B's temperatures vary much more from day to day than City A's. Therefore, they have the same mean but different variance and standard deviation.

5. Which City Has More Predictable Temperatures?

Answer: City A has more predictable temperatures because its variance is much smaller, meaning the temperatures stay close to the average.

Case Study 4-Data Centre CPU Usage

A data scientist records CPU usage percentages:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

1. Calculate mean, median, and mode.
2. Calculate the range.
3. Find Q1, Q3 and IQR.
4. Identify possible outliers.
5. Determine the direction of skewness.

Given Data (CPU Usage %):

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

1. Calculate Mean, Median, and Mode

(a) Mean

Formula:

$$\text{Mean} = \frac{\sum X}{N}$$

Calculation:

$$\begin{aligned}\sum X &= 40 + 42 + 45 + 45 + 46 + 48 + 50 + 52 + 55 + 90 = 513 \\ N &= 10 \\ \text{Mean} &= \frac{513}{10} = 51.3\end{aligned}$$

Mean = 51.3%

(b) Median

For an even number of observations,

$$\text{Median} = \frac{\left(\frac{N}{2}\right)^{th} + \left(\frac{N}{2} + 1\right)^{th}}{2}$$

Calculation:

Ordered data: 40, 42, 45, 45, 46, 48, 50, 52, 55, 90

$$\text{Median} = \frac{46 + 48}{2} = 47$$

Median = 47%

(c) Mode

Mode = Value occurring most frequently.

The value 45 occurs 2 times.

Mode = 45%

2. Calculate the Range

Formula: Range = Maximum – Minimum

Calculation: $90 - 40 = 50$

Range = 50%

3. Find Q1, Q3 and IQR

Formula: $IQR = Q_3 - Q_1$

Lower half: 40, 42, 45, 45, 46

$$Q_1 = 45$$

Upper half: 48, 50, 52, 55, 90

$$Q_3 = 52$$

$$IQR = 52 - 45 = 7$$

- $Q_1 = 45$
- $Q_3 = 52$
- $IQR = 7$

4. Identify Possible Outliers

Formula:

Lower Fence: $Q_1 - 1.5(IQR)$

Upper Fence: $Q_3 + 1.5(IQR)$

Calculation:

$$1.5 \times 7 = 10.5$$

Lower Fence: $45 - 10.5 = 34.5$

Upper Fence: $52 + 10.5 = 62.5$

Any value below 34.5 or above 62.5 is an outlier.

Since

$$90 > 62.5$$

Outlier = 90%

5. Determine the Direction of Skewness

The data contains a very large value (90), which pulls the mean to the right.

Also,

- Mean = 51.3
- Median = 47
- Mean > Median

Therefore, the distribution is positively skewed (right-skewed).

Case Study 5-Comparing Two Datasets

Two data analysts receive the following datasets:

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

1. Calculate the mean of both datasets.
2. Calculate the variance of both.
3. Calculate the standard deviation of both.
4. Which dataset has greater spread?
5. If both datasets represent model errors, which model appears more consistent?
6. Explain why mean alone is not enough to compare datasets.

Solution:

Given Data

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

1. Calculate the Mean

Formula:

$$\bar{x} = \frac{\sum X}{N}$$

Dataset A:

$$\begin{aligned}\sum X &= 10 + 20 + 30 + 40 + 50 = 150 \\ \bar{x} &= \frac{150}{5} = 30\end{aligned}$$

Mean of Dataset A = 30

Dataset B:

$$\sum X = 28 + 29 + 30 + 31 + 32 = 150$$
$$\bar{x} = \frac{150}{5} = 30$$

Mean of Dataset B = 30

2. Calculate the Variance

Formula (Population Variance):

$$\sigma^2 = \frac{\sum (X - \bar{x})^2}{N}$$

Dataset A:

X	$X - \bar{x}$	$(X - \bar{x})^2$
10	-20	400
20	-10	100
30	0	0
40	10	100
50	20	400

$$\sum (X - \bar{x})^2 = 1000$$
$$\sigma^2 = \frac{1000}{5} = 200$$

Variance of Dataset A = 200

Dataset B:

X	$X - \bar{x}$	$(X - \bar{x})^2$
28	-2	4
29	-1	1
30	0	0
31	1	1
32	2	4

$$\sum (X - \bar{x})^2 = 10$$

$$\sigma^2 = \frac{10}{5} = 2$$

Variance of Dataset B = 2

3. Calculate the Standard Deviation

Formula: $\sigma = \sqrt{\sigma^2}$

Dataset A:

$$\sigma = \sqrt{200} = 14.14$$

Standard Deviation = 14.14

Dataset B

$$\sigma = \sqrt{2} = 1.41$$

Standard Deviation = 1.41

4. Which Dataset Has Greater Spread?

Compare the standard deviations:

- Dataset A = 14.14
- Dataset B = 1.41

Answer: Dataset A has the greater spread because its values are much farther from the mean.

5. If Both Datasets Represent Model Errors, Which Model Appears More Consistent?

A smaller standard deviation means the errors are closer to the mean and more consistent.

Answer: Dataset B is more consistent because it has a much smaller standard deviation.

6. Why Is Mean Alone Not Enough to Compare Datasets?

The mean only shows the average value. It does not indicate how the data is distributed around the average.

In this example, both datasets have the same mean (30), but:

- Dataset A has values spread widely (10 to 50).
- Dataset B has values clustered closely (28 to 32).

Therefore, measures like variance and standard deviation are needed to understand the variability and consistency of the data.